

Marwan Badr

AI Agent Architect & Backend Engineer | Multi-Agent Systems, RAG & ERP

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portfolio · Cairo, Egypt



PROFESSIONAL SUMMARY

AI Agent Architect and Software Developer specializing in production **Multi-Agent Systems (LangGraph, StateGraph)**, **Advanced RAG pipelines**, and **Agentic Evaluation (Evals)** frameworks. Combines an accounting domain foundation (English Commerce) with production experience in Python, FastAPI, and vector search (Qdrant) to architect autonomous, guarded, and deterministic LLM agents with Human-in-the-Loop (HITL) safety controls. Proven track record building end-to-end multi-agent enterprise SaaS, native ERP integrations (Odoo, Frappe), and high-throughput real-time sync systems.

TECHNICAL SKILLS

AI & Agentic Systems	LangGraph (StateGraph, Checkpointing, Dynamic Routing), LangChain, Multi-Agent Architecture (Router & Desks), Deterministic Tool Calling, Structured Outputs (Pydantic), Human-in-the-Loop (HITL), Reflection & Self-Correction, Guardrails & Execution Policies
RAG & Vector Search	Advanced RAG, Hybrid Search (Dense + Sparse), Qdrant Vector DB, Semantic Caching, Chunking Strategies, Metadata Filtering, Context Optimization, Knowledge Base Indexing
LLM Evals & MLOps	Agent Evaluation (Evals, Accuracy Benchmarking, Ground Truth Datasets, LLM-as-a-Judge), Tracing & Telemetry (LangSmith, OpenTelemetry), Token Budgeting & Cost Metering, Model Routing (DeepSeek, Gemini, OpenAI, Claude), Prompt Engineering
Backend & API	Python, FastAPI (AsyncIO), Django, Django REST Framework (DRF), RESTful APIs, Webhooks, SSE Streaming, Celery, Redis
Odoo Development	Odoo 17, Odoo 18, Custom Modules, ORM, XML Views, Security Rules, In-ERP Web Client UI, APIs
Frappe & ERPNext	Frappe Framework (v14, v15), ERPNext Customization, DocTypes, Server/Client Scripts, Hooks, Jinja
Databases & Caching	PostgreSQL (Row-Level Security / RLS), MySQL, MariaDB, Redis (Caching & Task Queues)
DevOps & Tools	Docker, Git, GitHub, Nginx (Zero-Copy X-Accel), Linux / Bash, Systemd Sandboxing
Accounting Domain	Professional Financial Accounting (PFA), Chart of Accounts, General Ledger, AR/AP Reconciliations, 10-Tier Bank Reconciliation, Period Audits, Financial Logic Modeling
Spoken Languages	Arabic (Native), English (Fluent – English Commerce Section)

PROFESSIONAL EXPERIENCE

Founder & Lead AI Architect

Aug 2026 – Present

Razzyyn AI (Autonomous Accounting Virtual Finance Dept SaaS) – Remote

- **Multi-Agent System Architecture:** Architected and deployed the production multi-agent system with **LangGraph** and **FastAPI**, orchestrating a Master Router and 6 specialized accounting desks (Ask, Analyse, Reconcile, Audit, Record, Update) with StateGraph checkpointing and dynamic checklist planning.
- **RAG & Tool Execution:** Engineered **Advanced RAG** using **Qdrant** vector store for standards/laws retrieval, and deterministic **Tool Calling** with Pydantic structured output validation.
- **Evals, Guardrails & Safety:** Implemented **LLM Evaluation pipelines (Evals)**, semantic guardrails, dual-gated write policies (monetary limits, blocked accounts, batch caps), and interactive **Human-in-the-Loop (HITL)** approval cards.
- **Algorithmic Optimization & Cost Metering:** Built AST SQL query rewrites (**sqlglot**) isolating live posted records, 10-tier algorithmic bank reconciliation, and multi-model token cost metering (DeepSeek, Gemini).
- **End-to-End Stack Delivery:** Directed full engineering across native Frappe apps (v14/v15), native Odoo modules (v17/v18), React/Vite admin panel, and multilingual Next.js landing platform.

Software / Frappe Developer

Mar 2026 – Present

Showsatellite (Home Appliances Retail) – Remote (Bahrain)

- Lead custom **Frappe Framework & ERPNext** engineering, DocType modeling, whitelisted Python APIs, and automated business workflows.
- Engineered a real-time bidirectional synchronization engine connecting **ERPNext** and **OpenCart**, automating product catalogs, multi-tier pricing, stock levels, and images with zero variance.
- Built a centralized operations logistics dashboard managing delivery scheduling, real-time driver dispatching, and field technician ticketing.

Accountant

Feb 2026 – Mar 2026

Minicash Furniture Company – Onsite

- Managed daily accounting operations, journal entries, Chart of Accounts maintenance, and AR/AP reconciliations.

FEATURED PROJECTS

Autonomous AI & Multi-Agent Systems

- **Razyyn AI — Autonomous AI Accountant Platform (Founder & Architect):** Multi-agent SaaS (FastAPI, LangGraph, Qdrant, Redis) for ERPNext and Odoo. Orchestrates 6 specialized accounting desks, advanced RAG retrieval, automated period audits, 10-tier algorithmic bank reconciliation, and guarded transactional ERP writes.
- **Razyyn AI — Native Odoo Integration Modules (v17 & v18):** Custom Odoo addons embedding in-ERP chat UI, guarded write gateway, AST-based live row SQL filtering, and multi-company security.
- **Razyyn AI — Native Frappe Applications (v14 & v15):** Custom Frappe apps providing ERPNext chat UI, multi-agent modes (Ask, Analyse, Audit), attachment OCR processing, and granular write policy controls.

High-Throughput Backend & Web Architecture

- **File Frog — High-Throughput Document Processing & Conversion Engine:** High-efficiency single-server platform (FastAPI, Next.js, Redis, Huey, Nginx) featuring 64KB chunked memory streaming, system RAM/disk backpressure controls, segmented worker queues, sandboxed LibreOffice CLI conversions, Tesseract OCR, and zero-copy Nginx **X-Accel-Redirect** downloads.
- **Integrated E-Commerce & Enterprise Operations Platform:** Full-scale customer ordering platform integrated with ERPNext, automating sales workflows, Telegram notifications, and logistics dispatch management.
- **Real-Time Synchronization Engine: ERPNext ↔ OpenCart:** Production bidirectional sync engine automating product catalogs, dynamic pricing, inventory levels, and high-speed image transfers with zero variance.
- **Office Leaks — Professional Social Network Platform:** Corporate transparency social platform built with Django and DRF, featuring company workplace reviews, culture ratings, nested discussions, and secure JWT authentication.

ERP Custom Addons & Enterprise Workflow Automations

- **Odoo Sales Commission Management Module:** Custom Odoo 17 module automating sales team commission tracking, computation rules linked directly to invoice records, and commission payout audit wizards.
- **Order Cancellation & Return Automation Frappe App:** Automatic multi-document rollback engine for Frappe/ERPNext cancelling linked Payment Entries, Sales Orders, and Delivery Notes upon Credit Note submission with Serial Number reactivation.
- **Support Management Frappe App:** Comprehensive support ticketing and field service app for Frappe/ERPNext, managing maintenance appointment schedules, technician allocation, and customer requests.
- **Telegram Notification Frappe App:** Dynamic event-driven Frappe application sending real-time Telegram alerts with interactive inline action buttons for document approvals.

** Additional repositories, live demos, and video walkthroughs available on portfolio: my-portfolio-portfolio-iota.vercel.app*

EDUCATION

Bachelor of Commerce – Accounting (English Section)

Grade: *Very Good*

MILITARY STATUS

Permanently Exempted